SARA (Smart Assistive Reading Aid)

Results, benchmarks and latency · 14 October 2026

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This document collects every measured result of the project in one place: how accurate each function is, how fast it runs on the Raspberry Pi 5 and on the laptop, and what is not measured. Each number comes from a saved result file or from the benchmark report written on the Pi on 13 October 2026. Where a number was measured on a hot, throttled board, or only on the laptop, it says so.

1. Accuracy of each function

Table 1: Accuracy on data the models were not trained on

Function	Test data	Result
Taka recognition	2,282 held-out pictures made like the training pictures	Precision 0.995, recall 0.996, mAP@0.5 0.995, mAP@0.5:0.95 0.849
Taka recognition	1,536 photographs by other people (Bangla Money)	94.1% right, 17 wrong (hand-held detector, 0.75); earlier detector 89.6%, 50 wrong (91.5% at 0.25)
Taka recognition, notes held in a hand	1,127 NSTU-BDTAKA close-ups, one per photo	47.5% right (earlier detector 11.8%); Pi 8-bit: 33.5% (earlier 8.7%)
Unknown notes (one Taka, not a class)	101 Bangla Money photos	Announced as a value: 36.6% (earlier 22.8%); Pi 8.9% (earlier 5.0%)
Several notes in one picture	900 pictures of 1, 2 or 3 test-split notes on new backgrounds	Spoken total right 97.0% (earlier detector 99.2%); never too many counted
Counterfeit check, 1 view / 6 views	208 test notes, usual split, 3 training runs	98.2 ± 0.7% / 98.7 ± 0.3% (71.2 ± 11.4% at 1 view without prefix training)
Counterfeit check on unseen prints	222 test notes whose counterfeit prints are not in training	Network alone about 90%; with the watermark check 94.4% (1 view), 95.0% (6 views)
Safe spoken verdict	88 test counterfeits	0 passed as "likely genuine"; the glass never says "counterfeit"
Safe spoken verdict, genuine notes confirmed	Close-up views / whole-note photos by other people	57.5% / 0.4–2%
Watermark check (MobileNetV2, in the glass)	197 unseen-print notes	92.9% (AUC 0.976); 92.6% on all 222 with the learned watermark finder
Watermark check, DMWR (47,818 parameters)	Same 197 notes, 3 training runs	91.5 ± 0.3%; not significantly different from MobileNetV2
UV fibre check, with the colour condition	111 real UV photos of genuine notes; simulated fibre-free printed sheets	85.6% of genuine photos accepted; 88.6% of held-out simulated sheets rejected. No real counterfeit tested

Function	Test data	Result
Text reading	Rendered benchmark, test set (432 pictures)	Letter error 2.3% (Bangla 3.0%, English 1.5%) with the new recogniser; 2.5% before it; 9.3% before this work
Text reading	1,000 real word photos (ICDAR 2019 MLT)	Letter error 28.7% Bangla, 50.6% Latin script with the recogniser trained on real words (39.2% and 62.5% before); nothing read on 12% and 33%
Text reading, picture enlarged to 880 instead of 1,200 pixels	Rendered test set	3.15% (full-precision detector), 3.12% (8-bit): worse, so not adopted
Object detection, YOLOv8s	All 5,000 COCO val2017 photos	mAP@0.5 0.551; 0.521 through the glass path (0.493 before a fix)
Object detection, YOLOv8n / YOLOv8m	Same	0.453 / 0.607
Emotion	RAF-DB official test, 3,068 faces	86.5% (macro-F1 0.795)
Handwritten Bangla letters (fall-back)	38,608 characters from 305 unseen writers	90.0%
Voice commands (new, 14 October)	39 command phrases spoken by 3 synthetic voices (117 commands); not people	Carried out: 99.1% clean, 99.1% at 20 dB noise, 96.6% at 10 dB noise; never a wrong command
Voice commands, false commands	90 ordinary English sentences and 10 Bangla sentences per noise level; 60 s of noise alone	English: 0 / 1 / 1 of 90 (clean / 20 dB / 10 dB); Bangla: 0 / 0 / 2 of 10; noise alone: 0
Bangla voice (Piper), machine-listener check	6 sentences given to an online Bangla speech recogniser	2 exact, 2 with small differences, 2 with wrong words

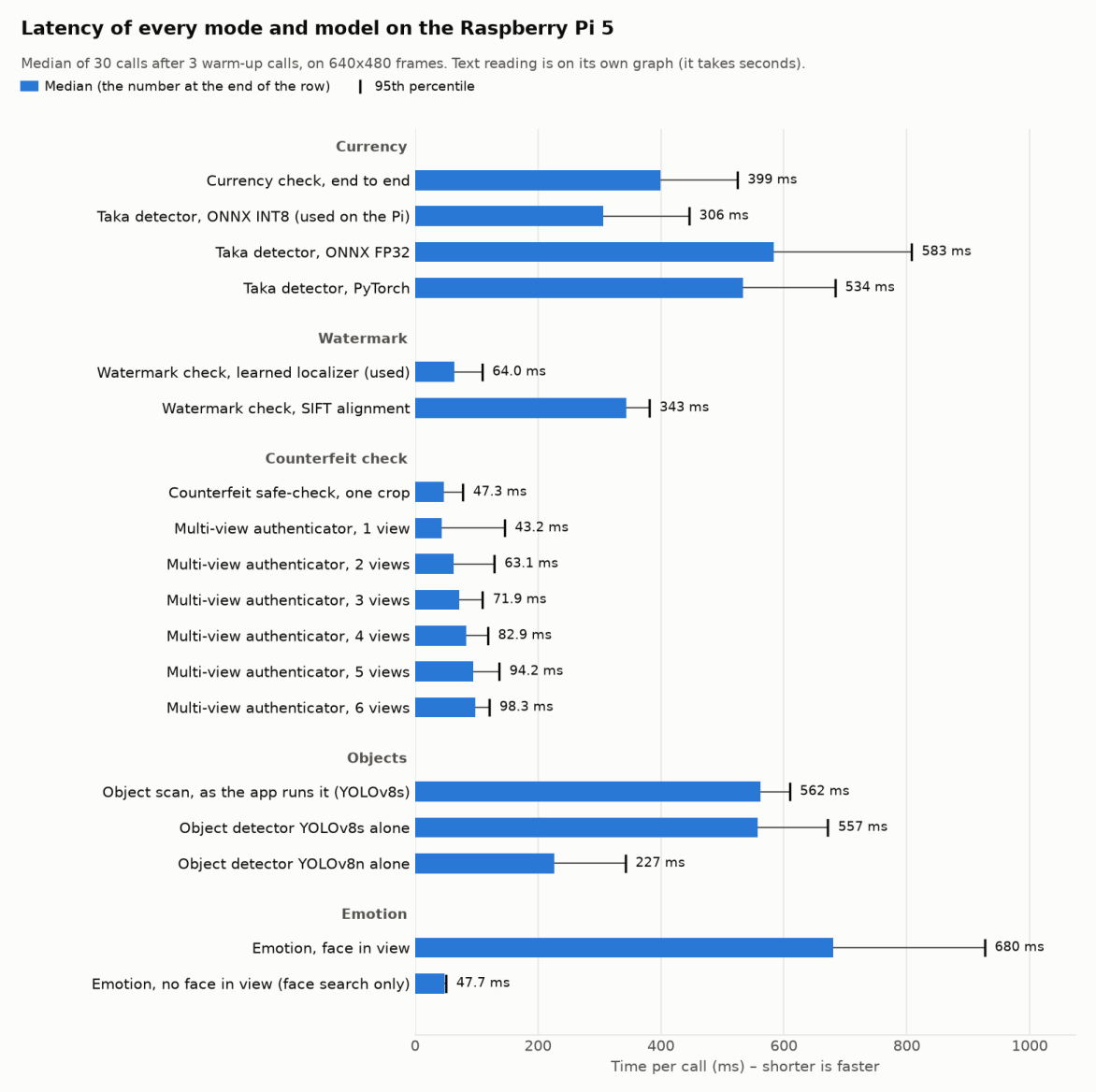
2. Latency on the Raspberry Pi 5

Two runs exist. The first (1 October 2026, 100 calls per row) ended at 77.1 °C with no throttling. The second (13 October 2026, 30 calls per row, after the code changes of 11–13 October) ran on a board that was already at 73 °C when it started, reached 88.7 °C, and was throttled or clock-capped in 76% of the samples of the text-reading test, with the CPU down to 1.5 GHz from 2.4 GHz. **The second run is therefore the speed of a hot glass.** How much of the difference between the runs is heat cannot be told from these two runs.

Table 2: Latency on the Raspberry Pi 5 (median; 95th percentile in brackets)

Part of the glass	13 October (hot board)	1 October
Currency check, end to end	399 ms (524 ms)	327 ms (560 ms)
Taka detector, INT8 ONNX (used on the Pi)	306 ms (446 ms)	254 ms (339 ms)
Taka detector, full-precision ONNX / PyTorch	583 ms / 534 ms	485 ms / 507 ms
Watermark check, learned finder (used)	64 ms (109 ms)	50 ms (62 ms)
Watermark check, SIFT alignment	343 ms (381 ms)	372 ms (854 ms)
Counterfeit safe-check, one crop	47 ms (78 ms)	36 ms

Part of the glass	13 October (hot board)	1 October
Counterfeit network, 1 view / 6 views	43 ms / 98 ms	37 ms / 90 ms
Object scan as the app runs it (YOLOv8s)	562 ms (610 ms)	not run
Object detector YOLOv8n alone	227 ms (342 ms)	207 ms (217 ms)
Emotion, face in view	680 ms (926 ms)	not run
Emotion, no face in view (face search only)	48 ms (50 ms)	48 ms (52 ms)



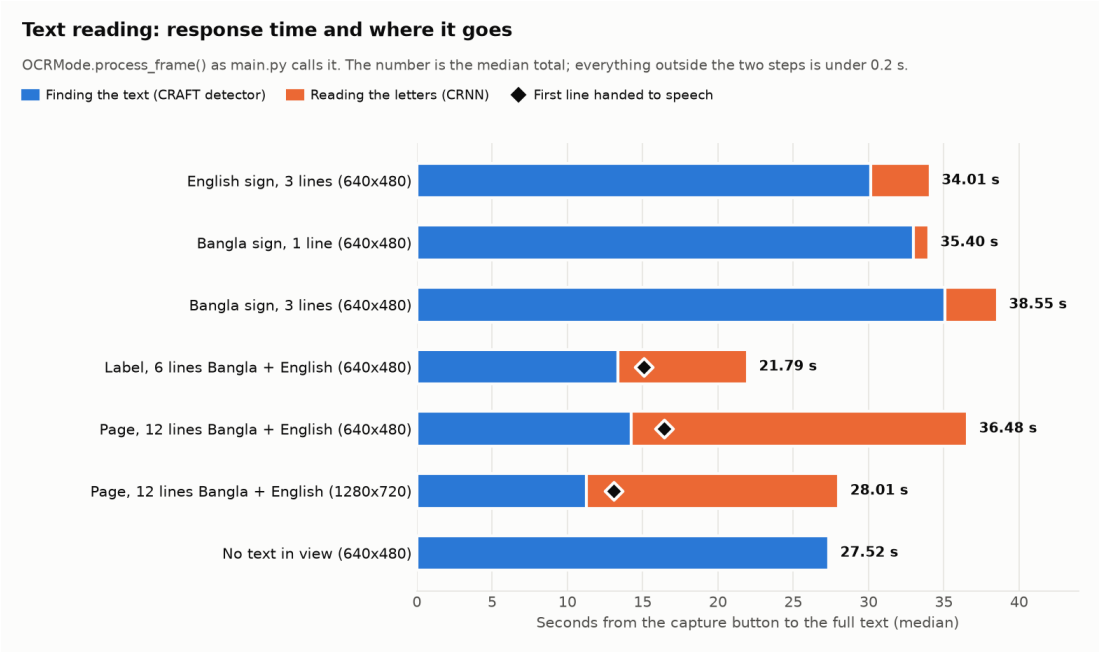
Latency of every mode and model on the Pi, 13 October 2026 (from the Pi's benchmark report).

Text reading on the Pi

Table 3: Text reading on the Raspberry Pi 5, 13 October 2026 (hot board; median of 3 to 5 captures)

Capture	Total	Finding the text	Reading the letters	First line spoken	Letter errors
English sign, 3 lines	34.0 s	30.1 s	4.0 s	–	0%
Bangla sign, 1 line	35.4 s	33.0 s	1.0 s	–	0%
Bangla sign, 3 lines	38.6 s	35.1 s	3.5 s	–	0%

Capture	Total	Finding the text	Reading the letters	First line spoken	Letter errors
Label, 6 lines, Bangla + English	21.8 s	13.3 s	8.6 s	15.1 s	1.0%
Page, 12 lines (640×480)	36.5 s	14.2 s	22.3 s	16.4 s	0%
Page, 12 lines (1280×720)	28.0 s	11.3 s	16.7 s	13.1 s	0%
No text in view	27.5 s	27.3 s	0 s	–	–
English sign, 3 lines, text found once (as on 1 October: 18.7 s)	18.3 s				



Text reading on the Pi: response time of each capture and where it goes (from the Pi's benchmark report).

What this run exposed, and what was changed on 14 October.

- **Short signs ran the text detector twice** (34.0 s against 18.3 s with the text found once). It now runs once per capture. On the laptop the new path reads the same text as before on 16 benchmark pictures. *Not yet timed on the Pi.*
- **A blank picture took 27.5 s and returned 48 made-up letters.** A picture with no text boxes now answers "no text found" after one detector pass. *Not yet timed on the Pi.*
- **One pass of the text detector costs 11 to 18 s on this board.** The 8-bit text detector is now the default on the Pi. It costs 0.57 points of letter error on the rendered validation set (3.15% against 2.58%), which is over the project's own 0.5-point rule; this is a stated exception made for speed. *Its speed on the Pi is not measured yet.*
- **The 1 October emotion figure (48 ms) was a frame without a face.** With a face, one answer takes 680 ms.

Other timings on the Pi, 13 October 2026

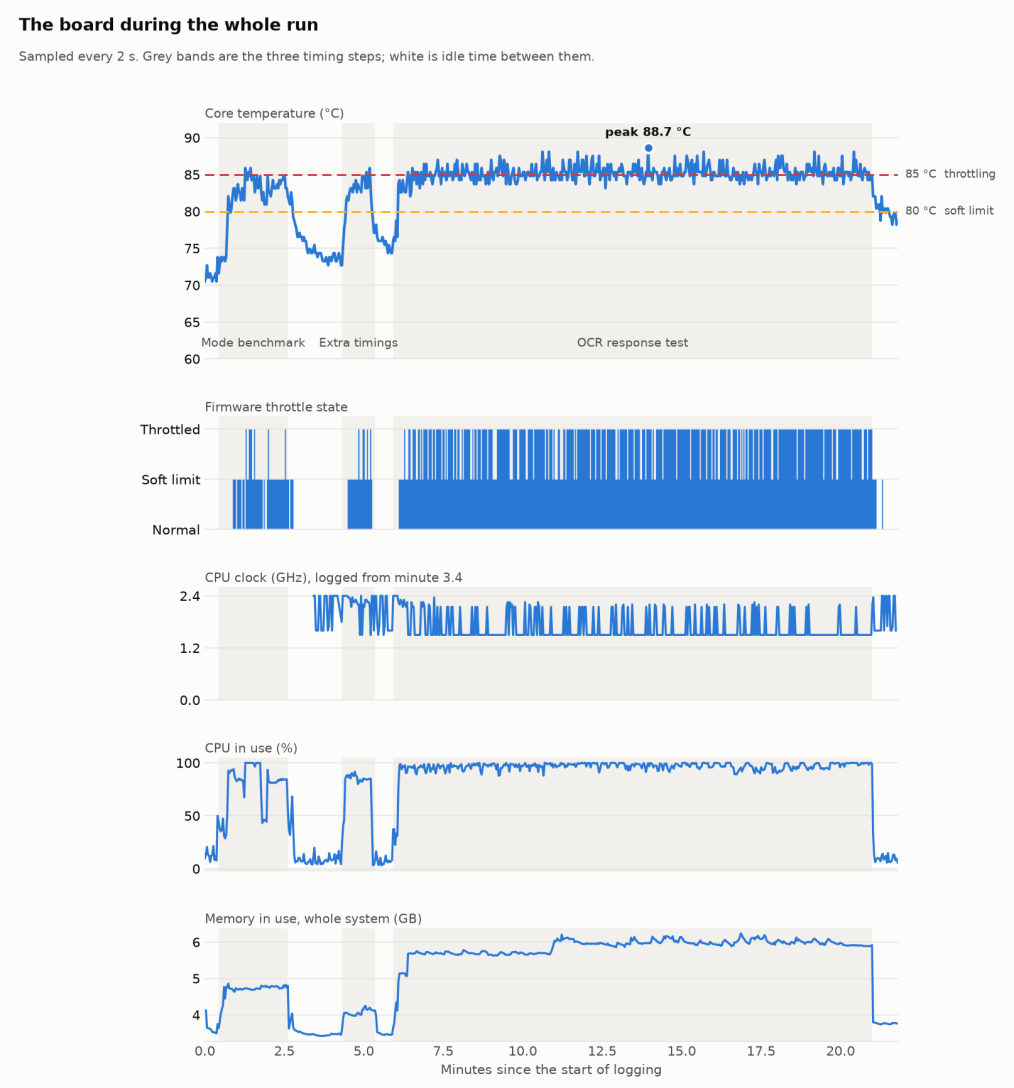
Table 4: Loading, speech, camera and the app's own log

What	Time
Loading the text reader (EasyOCR, Bangla + English)	8.7 s

What	Time
Currency mode, first switch into it / warm-up pass	2.1 s / 1.5 s
Emotion detector / object mode / watermark checker / counterfeit network, loading	2.2 s / 0.7 s / 0.6 s / 0.3 s
Camera, open to first frame	1.4 s
Camera, time between frames (640×480, indoors at night)	132 ms (7.6 frames a second)
Speech: making the audio for a short Bangla / English sentence	17.5 ms / 14.3 ms
App log, two sessions with the real camera: currency check (8)	406 ms median
App log: object scan and announcement (10)	777 ms median
App log: watermark check, guided, several frames (1)	3.6 s
App log: UV fibre check (3)	0.8 to 1.4 s
App log: one text capture	39.9 s
Memory: mode benchmark / text test / running app with all modes and preview	1.44 GB / 2.33 GB / 2.99 GB

Table 5: Board state during the runs

	13 October	1 October
Peak temperature	88.7 °C	77.1 °C (30 minutes of mixed use)
Throttled or clock-capped	76% of samples in the text test; 11% in the mode benchmark	never
Lowest CPU clock logged	1.5 GHz (full speed 2.4 GHz)	not logged
Under-voltage	never	never
Peak memory in the benchmark	1.44 GB	1.48 GB



Temperature, throttling and CPU clock during the 13 October run (from the Pi's benchmark report).

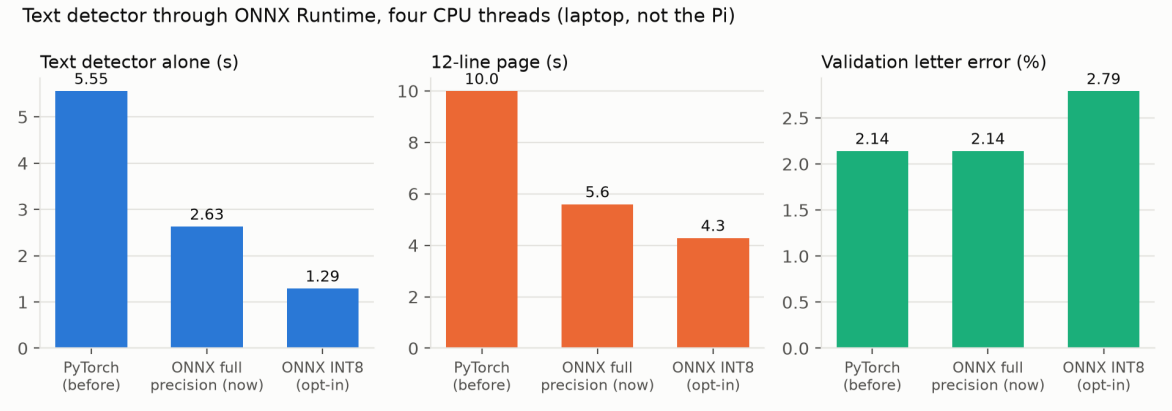
3. Speed measured on the laptop

These were measured on the development laptop (AMD Ryzen 7 5800H, RTX 3050), with the CPU limited to four threads where stated, the number of cores of the Pi. **They are not Pi numbers.** Only ratios can be expected to carry over.

Table 6: Laptop timings

What	Before	After	Accuracy
Text detector alone, 1,200×900 picture, 4 CPU threads (PyTorch → ONNX Runtime)	5.55 s	2.63 s	Identical output; validation error 2.14% both
12-line page, 4 CPU threads (same change)	9.99 s	5.59 s	Same
One benchmark picture, 4 CPU threads (same change)	2.30 s	1.22 s	Same
Text detector, 8-bit (INT8), 4 CPU threads	2.63 s	1.29 s (page 4.28 s)	Validation error +0.57 to +0.65 points
12-line page, beam search → greedy decoder (11 October run)	8.9 s	6.5 s	Validation error 2.18% → 2.14%

What	Before	After	Accuracy
12-line page spoken line by line: first line ready	nothing spoken until the page is done	about 5.5 s	Unchanged
Object scan through the glass path, 4 CPU threads (shrunk frame → frame as it is)	163 ms	173 ms	mAP@0.5 0.493 → 0.521
Speech: rendering one Bangla sentence of 1.5 to 4.3 s with the Piper voice (voice loaded)		0.2 to 0.5 s	Not timed on the Pi
Taka detector on the laptop GPU / CPU (PyTorch)		13.9 ms / 109 ms	
Taka detector, INT8 ONNX, laptop CPU		101 ms	
Counterfeit network, 6 views, laptop GPU		42.6 ms	
Watermark check, learned finder / SIFT, laptop CPU		15.9 ms / 97.9 ms	



Text detector through ONNX Runtime on four laptop CPU threads.

4. Results that corrected an earlier number

- **Objects.** 0.604 mAP came from 250 photos; on all 5,000 the same model scores 0.551.
- **Text on real photos.** 2.5% letter error on rendered text; 39% (Bangla) and 62% (Latin script) on real street words.
- **UV check.** The first design accepted 92.8% of genuine photos, and in a simulation also 93% of fibre-free printed sheets. With a colour condition: 85.6% of genuine photos accepted, about 89% of simulated sheets rejected.
- **Counterfeit benchmark.** About 98% on the usual split; about 90% on prints never seen in training (95.0% with the watermark).
- **Emotion on the Pi.** 48 ms was a frame without a face; with a face it is 680 ms.

5. Not measured

- The Pi after the 14 October changes: single detector pass, blank-picture answer, 8-bit text detector.
- The rebuilt voice on the Pi: how it sounds, and how long the Piper voices take to render there.

- Voice commands spoken by people, with the glass's microphone, in a room or a street.
- The Pi on a cool board with the current code, and the 30-minute sustained run with it.
- Any counterfeit note under the UV lamp; any function through the glass camera in daylight.
- The online description mode (the account has no credit).
- Battery life.
- A study with visually impaired users (trial sessions have started; no result is reported).

Sources. Pi figures: benchmark_report/REPORT.md and its raw files (13 October), and the 1 October run in the thesis. Laptop and accuracy figures: the result files under savior_glass/results/ and paper_evidence/.